

MSC-IT Modern Networking Practical Writeups

Practical No. 1: Configure IP SLA Tracking and Path Control Topology

Aim

To configure IP SLA tracking and path control topology using routers in GNS3.

Theory

IP SLA (Internet Protocol Service Level Agreement) is a Cisco feature used to monitor network performance such as delay, jitter, and connectivity. It helps administrators track the availability of a network path and perform automatic path control. Static and dynamic routing protocols are used to maintain communication between routers.

Procedure

1. Configure three routers with serial interfaces and loopback addresses.
2. Assign IP addresses to all interfaces.
3. Configure static routing on Router 1.
4. Configure EIGRP routing on Router 2 and Router 3.
5. Verify connectivity using ping commands.
6. Configure hostnames for routers.
7. Configure IP SLA probes on Router 1.
8. Schedule IP SLA operation and verify configuration.

Result

IP SLA tracking and path control topology was successfully configured and verified using ping and routing commands.

Conclusion

The practical demonstrated how IP SLA can be used to monitor network connectivity and improve path control in a routed network.

Practical No. 2: Implementation of BGP using AS_PATH Attribute

Aim

To implement Border Gateway Protocol (BGP) using the AS_PATH attribute.

Theory

BGP is a dynamic routing protocol used for routing between autonomous systems. The AS_PATH attribute is used to identify the route path and prevent routing loops. BGP selects the best path based on routing attributes.

Procedure

1. Configure IP addresses on three routers.
2. Configure loopback interfaces on all routers.
3. Enable BGP on all routers with different autonomous system numbers.
4. Configure neighbor relationships between routers.
5. Advertise loopback networks using network statements.
6. Verify routing table using show IP route command.
7. Test connectivity using ping commands.

Result

BGP was successfully configured and routes were exchanged using AS_PATH attributes.

Conclusion

The practical helped in understanding BGP configuration and the importance of the AS_PATH attribute in routing decisions.

Practical No. 3: Configuring IBGP and EBGp Sessions

Aim

To configure Internal BGP (IBGP) and External BGP (EBGP) sessions.

Theory

IBGP is used for communication between routers within the same autonomous system, whereas EBGp is used between routers in different autonomous systems. OSPF is commonly used as an interior routing protocol along with BGP.

Procedure

1. Configure IP addresses on all router interfaces.
2. Verify interface status.
3. Configure OSPF routing in Autonomous System 65200.
4. Configure BGP on Router 1, Router 2, and Router 3.
5. Configure neighbor relationships for IBGP and EBGp sessions.
6. Redistribute OSPF routes into BGP.
7. Verify routes and test connectivity using ping commands.

Result

IBGP and EBGp sessions were successfully established and routes were exchanged correctly.

Conclusion

The practical demonstrated the configuration and operation of IBGP and EBGP sessions in multi-router environments.

Practical No. 4: Secure Management Plane

Aim

To configure and secure the management plane of routers.

Theory

The management plane is responsible for administrative access to networking devices. Security measures such as passwords, encryption, banners, and Telnet authentication help protect routers from unauthorized access.

Procedure

1. Configure IP addresses on all routers.
2. Configure static routing between routers.
3. Verify connectivity using ping commands.
4. Configure hostname on routers.
5. Set minimum password length and enable secret password.
6. Configure console, VTY, and AUX line passwords.
7. Enable password encryption.
8. Configure MOTD banner.
9. Test remote access using Telnet.

Result

Secure management access was successfully configured and verified on routers.

Conclusion

The practical demonstrated how to secure router management access using passwords, encryption, and Telnet authentication.

Practical No. 5: Configure and Verify Path Control using PBR

Aim

To configure and verify path control using Policy Based Routing (PBR).

Theory

Policy Based Routing allows routers to make forwarding decisions based on policies instead of routing tables alone. Access lists and route maps are used to define custom routing paths.

Procedure

1. Configure IP addresses and loopback interfaces on four routers.
2. Configure EIGRP routing on all routers.
3. Verify routing tables and connectivity.
4. Configure access lists for source traffic.
5. Configure route maps to define alternate paths.
6. Apply policy routing on interfaces.
7. Verify traffic path using traceroute commands.

Result

Policy Based Routing was successfully configured and traffic followed the specified paths.

Conclusion

The practical demonstrated how PBR provides flexible traffic forwarding and path control in a network.

Practical No. 6: Demonstrate Inter VLAN Routing

Aim

To demonstrate inter VLAN routing using switches and routers.

Theory

Inter VLAN routing allows communication between devices in different VLANs. VLANs improve network segmentation, while trunking and router-on-a-stick configuration enable routing between VLANs.

Procedure

1. Check VLAN configuration on all switches.
2. Disable unused ports.
3. Configure basic switch settings such as hostname and passwords.
4. Configure access ports on switches.
5. Configure IP addresses on PCs and server.
6. Configure VTP server and client modes.
7. Configure trunk links between switches.
8. Create VLANs and assign VLAN names.
9. Configure VLAN interfaces on switches.
10. Assign switch ports to VLANs.
11. Configure router subinterfaces for inter VLAN routing.
12. Verify connectivity between VLANs.

Result

Inter VLAN routing was successfully configured and devices in different VLANs communicated with each other.

Conclusion

The practical demonstrated VLAN configuration, trunking, VTP, and router-on-a-stick inter VLAN routing.

Practical No. 7: Simulating MPLS Environment

Aim

To simulate an MPLS environment using routers.

Theory

MPLS (Multiprotocol Label Switching) improves packet forwarding efficiency using labels instead of IP lookups. LDP and MP-BGP are important protocols used in MPLS networks.

Procedure

1. Configure IP addressing on MPLS core routers.
2. Configure OSPF routing between routers.
3. Verify loopback connectivity.
4. Configure LDP on MPLS core interfaces.
5. Verify MPLS interfaces and LDP neighbors.
6. Configure MP-BGP sessions between routers.
7. Verify VPNv4 BGP sessions.
8. Add routers and configure VRFs.
9. Verify MPLS routing and connectivity.

Result

The MPLS environment was successfully simulated and MPLS label switching was verified.

Conclusion

The practical demonstrated MPLS configuration using LDP, OSPF, MP-BGP, and VRF technologies.

Practical No. 8: Simulating VRF (Virtual Routing and Forwarding)

Aim

To simulate VRF (Virtual Routing and Forwarding) in an MPLS network.

Theory

VRF allows multiple routing tables to exist on the same router. It enables network segmentation and secure routing for different customers in MPLS networks.

Procedure

1. Create VRF on Router 1.
2. Assign interfaces to the VRF.
3. Reconfigure IP addresses on VRF interfaces.
4. Verify VRF routing tables.
5. Configure OSPF routing inside the VRF.
6. Repeat VRF configuration on additional routers.
7. Redistribute OSPF routes into MP-BGP.
8. Redistribute MP-BGP routes back into OSPF.
9. Verify routing tables and connectivity.
10. Test end-to-end communication using ping and traceroute.

Result

VRF was successfully configured and end-to-end MPLS connectivity was achieved.

Conclusion

The practical demonstrated how VRF provides separate routing tables and secure traffic isolation in MPLS networks.